

Final Project: Chat Application

Objective:

Build a desktop chat application using Python that combines networking, GUI, and media processing.

Required Features:

- Socket programming using TCP (Client + Server)
- Send Text Messages
- Send Images (with compression before sending)
- Add hd button when activated by the user, send image without compression like WhatsApp
- Send Files / Videos
- GUI using Tkinter
- Chat UI with messages aligned (left/right like real chat apps)

Advanced Features (Mandatory):

- Host the application online using ngrok or Railway or AWS etc.....

Optional Features:

- Add camera capture and send camera photos
- Add voice recording and send audio
- Add progress bar while sending files
- Add the date when the message was sent

Team Members:

Number of team members is

- 2-members maximum 3
- 4-members are allowed if optional features are added
- All members prefer to participate in GitHub

Tricky Constraints (AI-resistant)

- You are NOT allowed to send file size directly before transfer
- You must design your own protocol to detect end of file
- You cannot use ready-made libraries for file transfer

- Each student must implement a UNIQUE variation (decided during evaluation)
- Code will be modified live during evaluation
- Code will be Checked and tested to see if it generated by Ai
- AI-generated code will receive zero.

Intentional Debug Task

You will receive a buggy code snippet during evaluation.

You must:

- Find the bug
- Explain why it happens
- Fix it live

Important Instructions:

- AI may be used in simple matters, such as using small functions or simple tasks that are not understood and do not require the full code.
- You must write your own code. Copying from AI tools or other students is not allowed.
- Your code will be tested with modifications and questions during evaluation.
- Any identical submissions or AI-generated code without understanding will receive zero.

